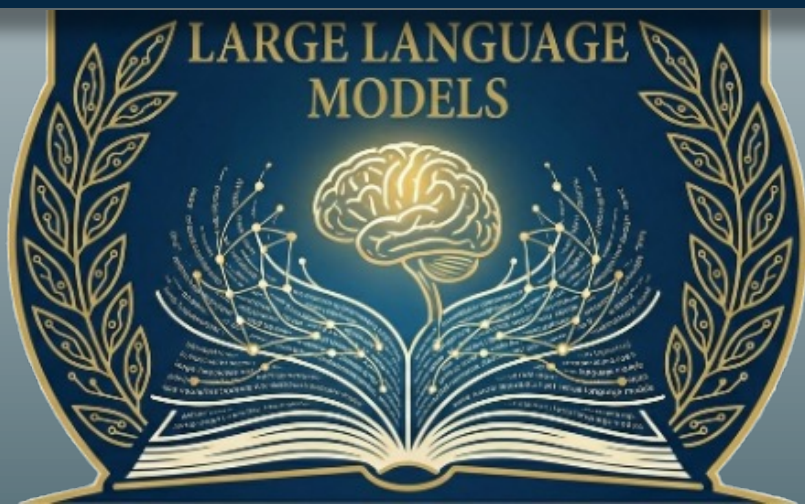


# Introduction



Mohammad Javad Dousti

# Instructor Info

- ❑ Mohammad Javad Dousti
- ❑ **Office:** ECE Building 2; Room 608
- ❑ **Office Hours**
  - Saturdays and Mondays 15:00 – 16:00
- ❑ **Email:** [mjdousti@ut.ac.ir](mailto:mjdousti@ut.ac.ir) (I'll try my best to respond within two business days)

# Class Info

- ❑ When: Saturdays & Mondays 16:30 – 18:00
- ❑ Where: Class #202
- ❑ Class web page:
  - E-Learn: <http://elearn.ut.ac.ir>
  - Discussion group:
    - Telegram group: <https://t.me/+thMUIQemsMU0MTRh>

# My introduction

- ❑ 1389, BSc in Computer Engineering from Sharif University of Technology
- ❑ 1394, MSc in Electrical Engineering from University of Southern California
- ❑ 1394, PhD in Computer Engineering from University of Southern California
- ❑ 1394–1399, Senior Research Scientist at Meta, Inc. (formerly Facebook, Inc.)
- ❑ 1399–Now, Assistant Professor at University of Tehran

What are large language models (LLMs)?

# Large language models

- They are large models of language
- What is (natural) **language**?
  - Built by humans to communicate

# Language: the game changer for humans



## Beyond First

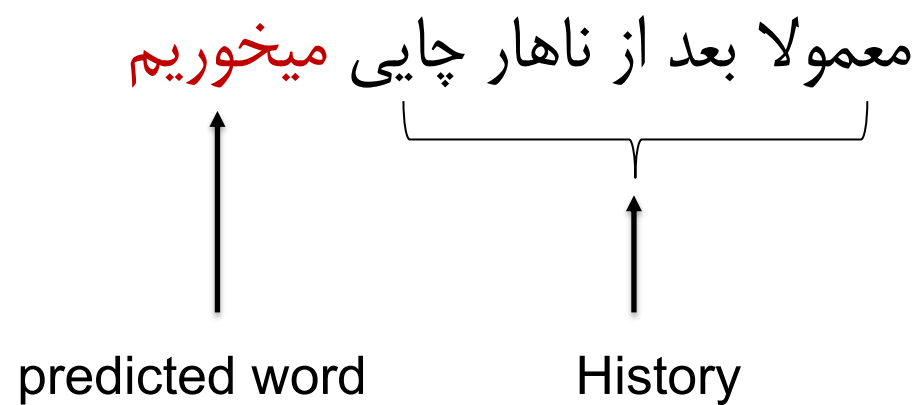
Language allows humans to communicate indirectly, without direct contact, overcoming spatial and temporal barriers.



## ary Advantage

Communication enables information exchange between individuals and generations, enhancing human adaptability and survival.

# Language modeling



| Model                             | Text                                                                                                                                                                                                                                                                                                               |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3-gram model trained on Moby Dick | <i>The meaning of life is</i> gulped and gone, bleeds with keenest anguish at the same muteness of humanity over all creations. Just so with Ahab, 'says I; look here you sabbee me, has every reason to be served.                                                                                                |
| RNN from [67]                     | <i>The meaning of life is</i> the tradition of the ancient human reproduction: it is less favorable to the good boy for when to remove her bigger. In the show's agreement unanimously resurfaced.                                                                                                                 |
| GPT-2 [68]                        | <i>The meaning of life is</i> something simple that has nothing to do with who we are. Everything about being with people is tied to those characteristics, and people are just machines.                                                                                                                          |
| GPT-3 davinci-002 [69]            | <i>The meaning of life is</i> a mystery to us all, and a question that will never be answered. Life is what we make of it, and it's up to each of us to find our own meaning in life.                                                                                                                              |
| GPT-4 (via ChatGPT interface)     | <i>The meaning of life is</i> subjective and varies greatly depending on personal beliefs, values, and experiences. Some people find meaning in relationships, personal achievements, or spiritual beliefs, while others may see it as a journey of self-discovery, learning, or contributing to the greater good. |

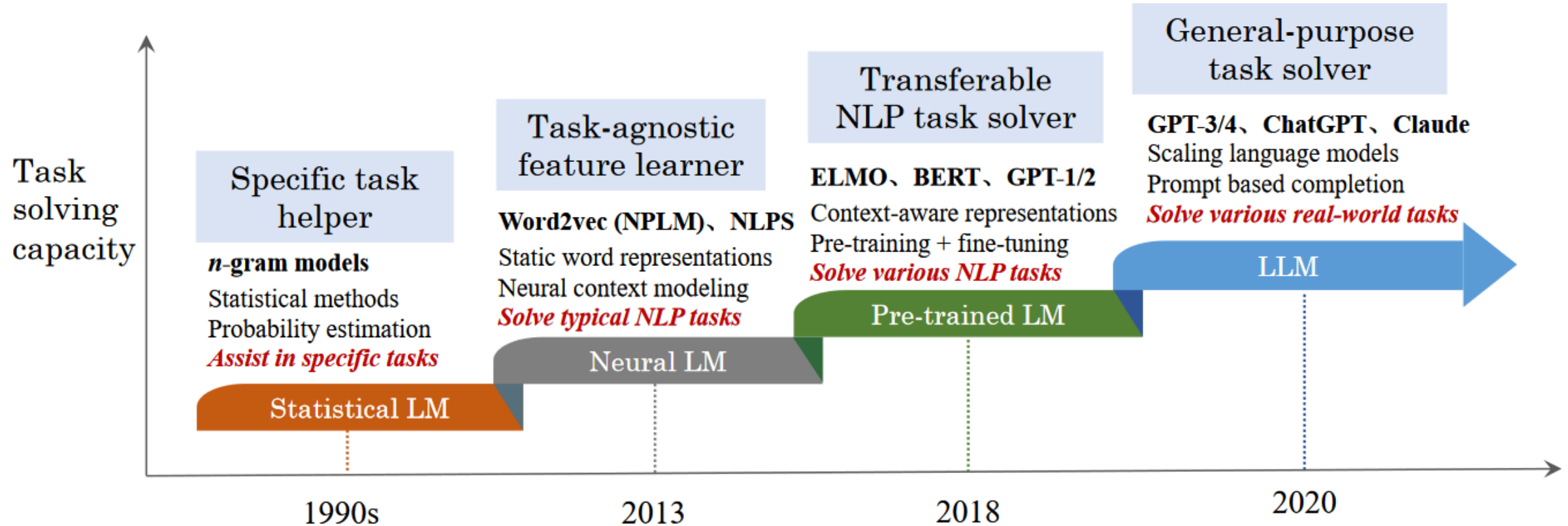
در مورد اهمیت مدل‌های زبانی بزرگ یک پاراگراف بنویس



Show thinking ▼

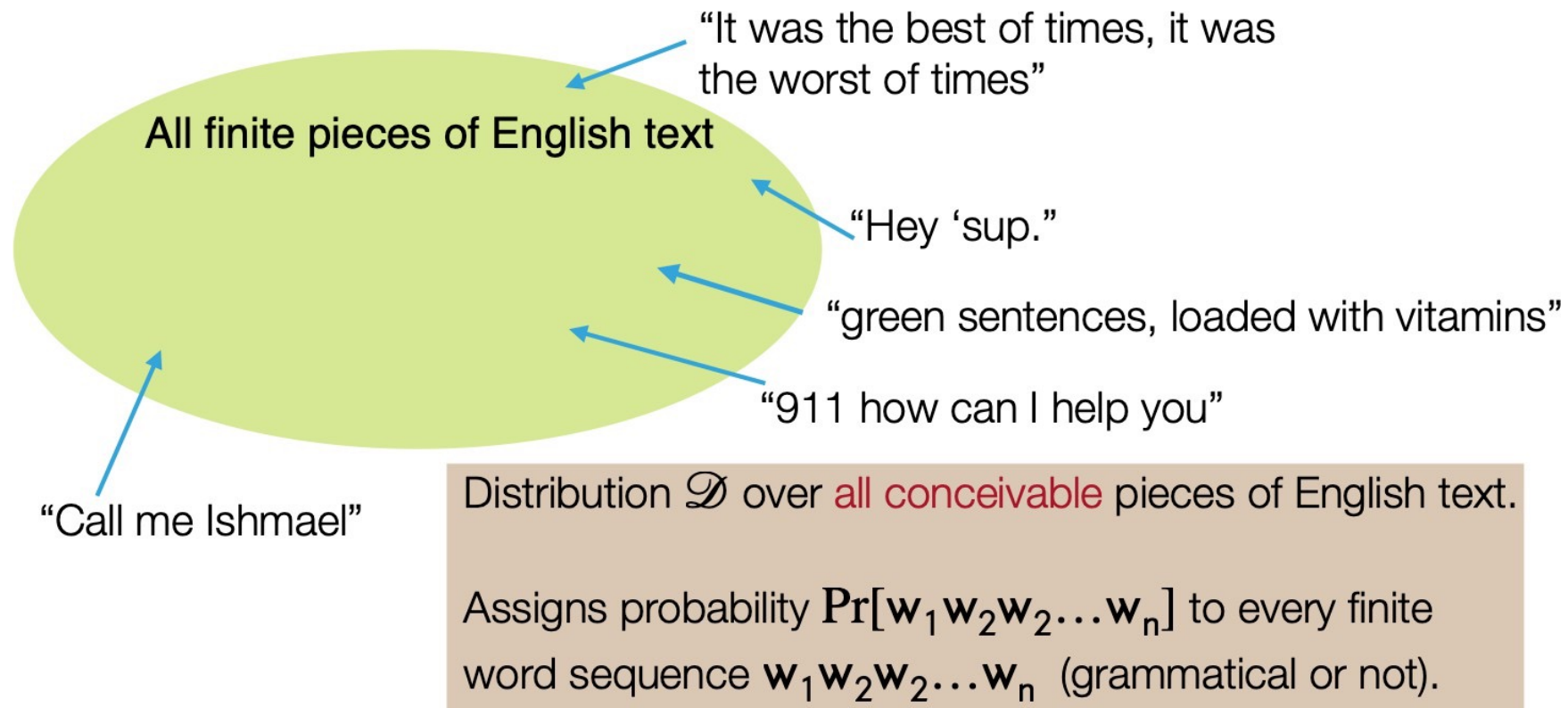
مدل‌های زبانی بزرگ (LLMs) به دلیل توانایی بی‌نظیرشان در درک، پردازش و تولید زبان طبیعی انسان، به هسته مرکزی انقلاب هوش مصنوعی در دنیای امروز تبدیل شده‌اند. این مدل‌ها با تحلیل حجم عظیمی از داده‌ها می‌توانند وظایف پیچیده شناختی مانند خلاصه‌سازی متون، ترجمه زبان‌ها، تولید کدهای برنامه‌نویسی و پاسخگویی به سوالات تحلیلی را با سرعت و دقتی خیره‌کننده انجام دهند. اهمیت واقعی آن‌ها در دگرگون‌سازی صنایع مختلف نهفته است؛ چرا که با خودکارسازی فرآیندها، شخصی‌سازی خدمات در حوزه‌هایی نظیر آموزش، پزشکی و تجارت، و ارتقای چشمگیر بهره‌وری انسانی، مرزهای تعامل انسان و ماشین را بازتعریف کرده و بستری قدرتمند برای نوآوری‌های آینده فراهم می‌آورند.

# Some history



# Language models: narrow sense

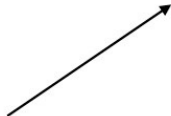
- A probabilistic model that assigns a probability  $P[w_1, w_2, \dots, w_n]$  to every finite sequence  $w_1, \dots, w_n$  (grammatical or not)



# Language models: narrow sense

$$p(w_1, w_2, w_3, \dots, w_N) = p(w_1) p(w_2|w_1) p(w_3|w_1, w_2) \times \dots \times p(w_N|w_1, w_2, \dots, w_{N-1})$$

Conditional probability



Sentence: “the cat sat on the mat”

$$\begin{aligned} P(\text{the cat sat on the mat}) &= P(\text{the}) * P(\text{cat}|\text{the}) * P(\text{sat}|\text{the cat}) \\ &\quad * P(\text{on}|\text{the cat sat}) * P(\text{the}|\text{the cat sat on}) \\ &\quad * P(\text{mat}|\text{the cat sat on the}) \end{aligned}$$

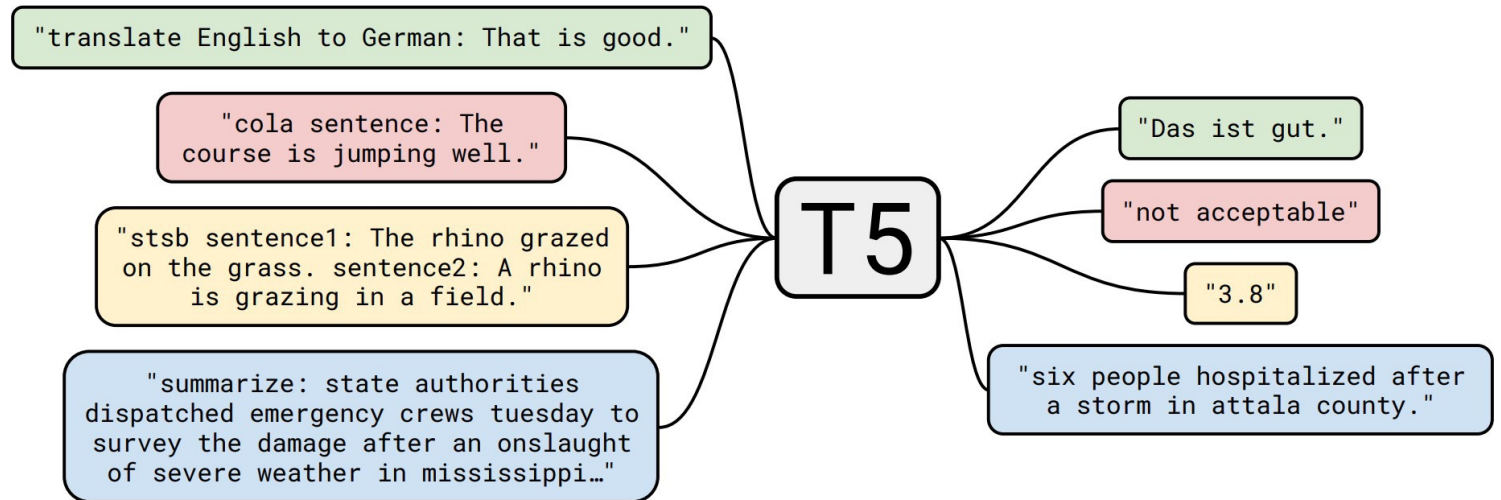
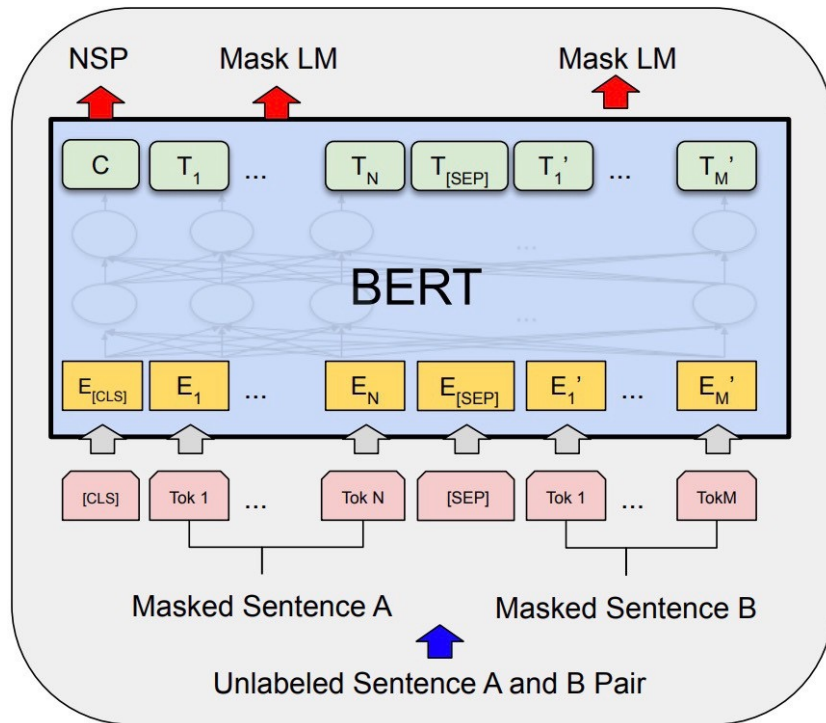
Implicit order



GPT-3 still acts in this way but the model is implemented as a very large neural network of 175-billion parameters!

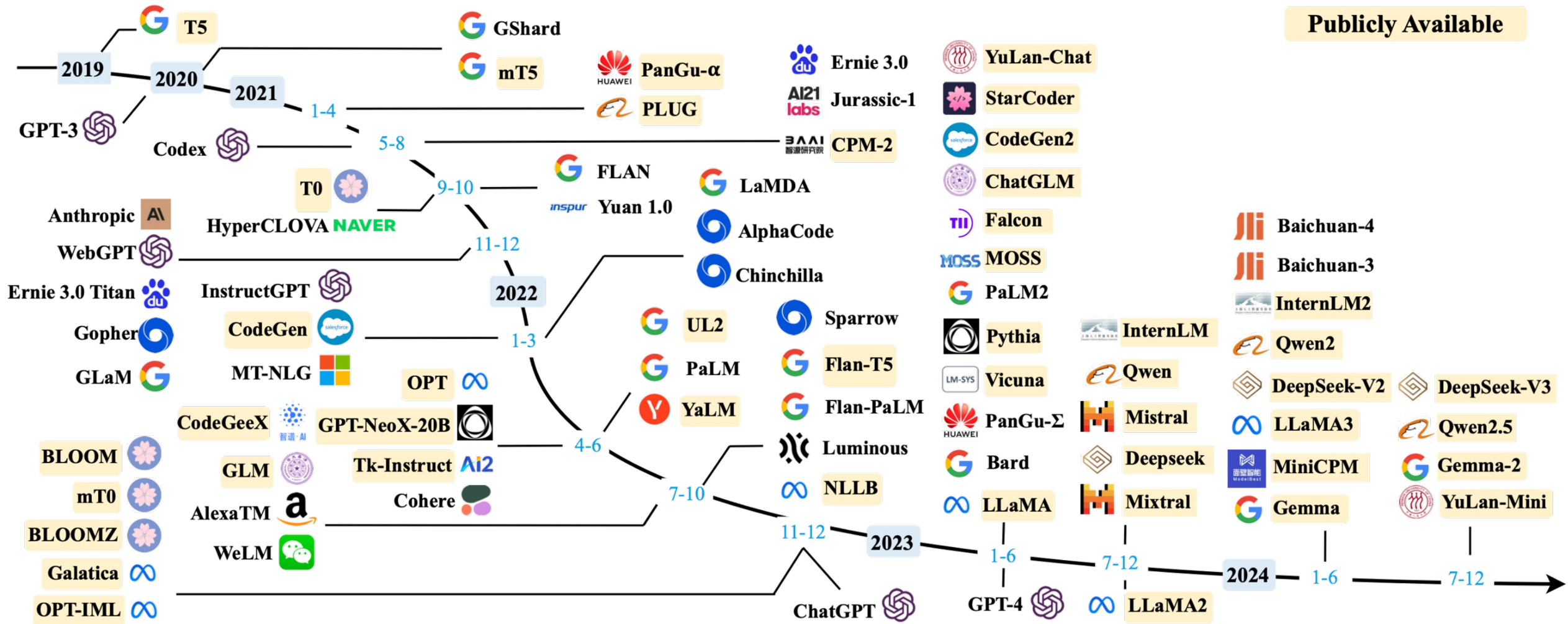
# Language models: broad sense

- ❑ Decoder-only models (GPT-x models)
- ❑ Encoder-only models (BERT, RoBERTa, ELECTRA)
- ❑ Encoder-decoder models (T5, BART)



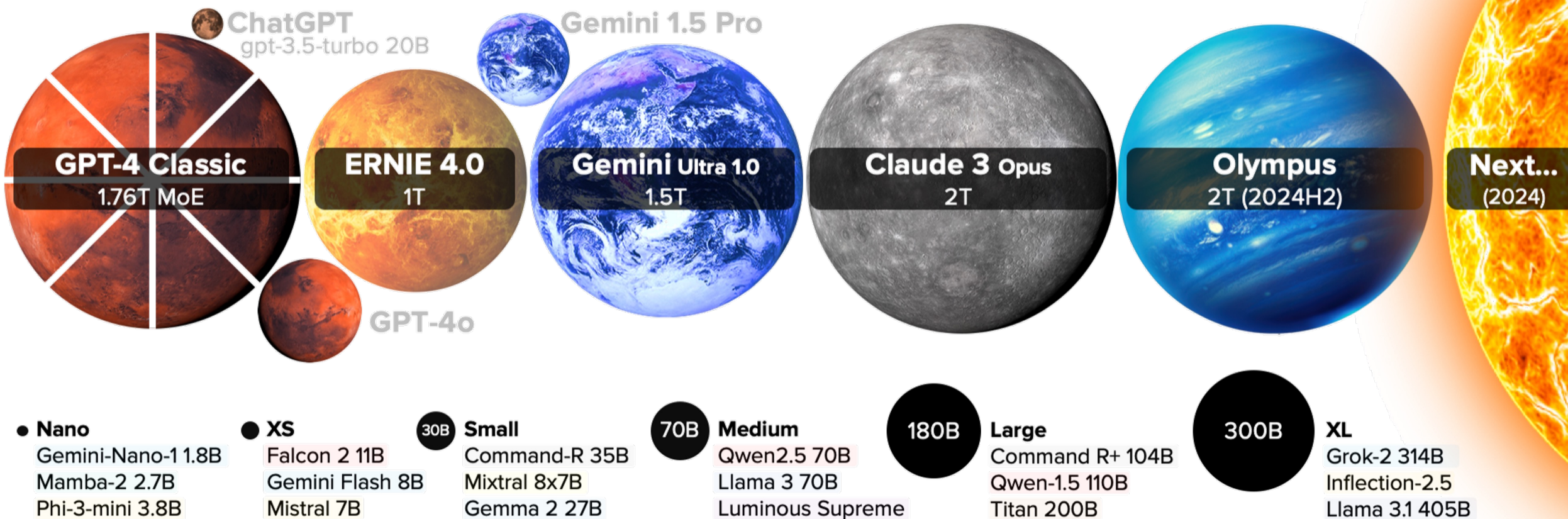
# How large are LLMs?

# Large Language Models (>10B Parameters)



W. X. Zhao et al. "A survey of large language models." *arXiv preprint arXiv:2303.18223* (2025).

# LARGE LANGUAGE MODEL HIGHLIGHTS (OCT/2024)



Parameters

AI lab/group

Sizes linear to scale. Selected highlights only. All 450+ models: <https://lifearchitected.ai/models-table/> Alan D. Thompson. 2021-2024.

Source: <https://lifearchitected.ai/models>

# Why LLMs?

- The promise: one single model to solve many NLP tasks

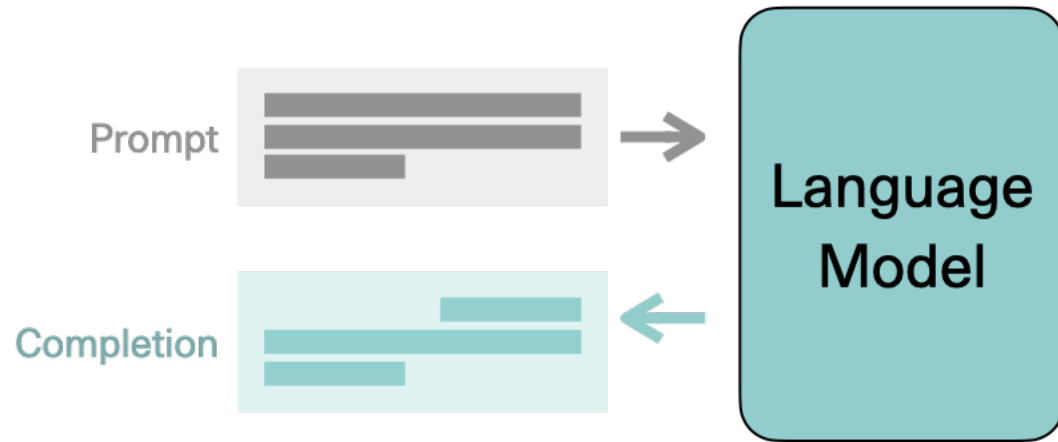
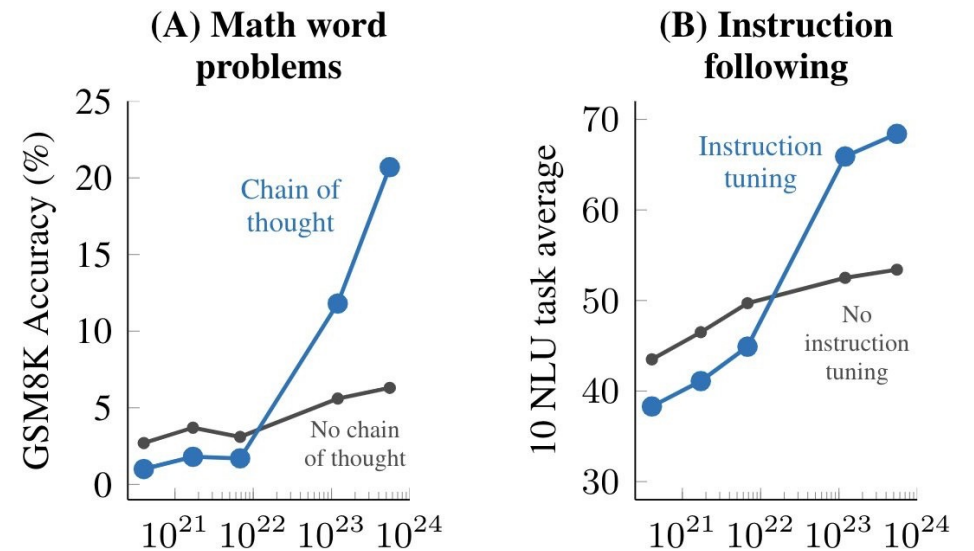


Image credit: Jay Alammar

- Emergent properties in LLMs



(Wei et al., 2022)

# Training Resource Requirement

| Model               | Training end | Chip type | TFLOP/s (max) | Chip count | Wall clock (days) | Total time (years) | Retail (US\$) | MMLU |
|---------------------|--------------|-----------|---------------|------------|-------------------|--------------------|---------------|------|
| <b>GPT-3 175B</b>   | Apr/2020     | V100      | 130           | 10,000     | 15 days           | 405y               | \$9M          | 43.9 |
| <b>Llama 1 65B</b>  | Jan/2023     | A100      | 312           | 2,048      | 21 days           | 118y               | \$4M          | 63.4 |
| <b>Llama 2 70B</b>  | Jun/2023     | A100      | 312           | 2,048      | 35 days           | 196y               | \$7M          | 68.0 |
| <b>Titan 200B</b>   | Apr/2023     | A100      | 312           | 13,760     | 48 days           | 1,319y             | \$45M         | 70.4 |
| <b>GPT-4 1.7T</b>   | Aug/2022     | A100      | 312           | 25,000     | 95 days           | 6,507y             | \$224M        | 86.4 |
| <b>Gemini</b>       | Nov/2023     | TPUv4     | 275           | 57,000     | 100 days          | 15,616y            | \$440M        | 90.0 |
| <b>Llama 3 405B</b> | Apr/2024     | H100      | 989           | 24,576     | 50 days           | 3,366y             | \$125M        | 85+  |
| <b>GPT-5</b>        | Apr/2024     | H100      | 989           | 50,000     | 120 days          | 16,438y            | \$612M        |      |
| <b>Grok 2</b>       | Jun/2024     | H100      | 989           | 20,000     | 50 days           | 6,571y             | \$245M        |      |
| <b>Olympus</b>      | Aug/2024     | H100      | 989           |            |                   |                    |               |      |
| <b>Gemini 2</b>     | Nov/2024     | TPUv6     | 1,847         |            |                   |                    |               |      |
| <b>Grok 3</b>       | Dec/2024     | H100      | 989           | 100,000    | 50 days           | 32,855y            | \$1.2B        |      |

Source: <https://lifearchitect.ai/models/>

# Why do we offer this course?

- ❑ **Problem:** Researchers are becoming disconnected from the underlying technology.
  - 9 years ago, researchers would implement and train their own models.
  - 7 years ago, researchers would download a model (e.g., BERT) and fine-tune it.
- ❑ Today, researchers just prompt a proprietary model (e.g., GPT-5/Claude/Gemini).
- ❑ Moving up levels of abstractions boosts productivity, but
  - These abstractions are leaky (in contrast to programming languages or operating systems).
  - There is still fundamental research to be done that require tearing up the stack.
- ❑ Full understanding of this technology is necessary for **fundamental research**.

# The industrialization of language models (1)

- ❑ GPT-4 supposedly has 1.8T parameters.
- ❑ GPT-4 supposedly cost \$100M to train.
- ❑ xAI builds cluster with 200,000 H100s to train Grok.
- ❑ Stargate (OpenAI, NVIDIA, Oracle) invests \$500B over 4 years.
- ❑ Meta Llama 5 will likely be closed source.
- ❑ Also, there are no public details on how frontier models are built.

# The industrialization of language models (2)

## 2 Scope and Limitations of this Technical Report

This report focuses on the capabilities, limitations, and safety properties of GPT-4. GPT-4 is a Transformer-style model [39] pre-trained to predict the next token in a document, using both publicly available data (such as internet data) and data licensed from third-party providers. The model was then fine-tuned using Reinforcement Learning from Human Feedback (RLHF) [40]. Given both the competitive landscape and the safety implications of large-scale models like GPT-4, this report contains no further details about the architecture (including model size), hardware, training compute, dataset construction, training method, or similar.

We are committed to independent auditing of our technologies, and shared some initial steps and ideas in this area in the system card accompanying this release.<sup>2</sup> We plan to make further technical details available to additional third parties who can advise us on how to weigh the competitive and safety considerations above against the scientific value of further transparency.

GPT-4 Technical Paper

# What can we learn from this course?

- ❑ There are three types of knowledge:
  - **Mechanics:** How things work (what a Transformer is, how model parallelism leverages GPUs)
  - **Mindset:** Squeezing the most out of the hardware, taking scale seriously (scaling laws)
  - **Intuitions:** Which data and modeling decisions yield good accuracy
- ❑ We can teach mechanics and mindset (these do transfer).
- ❑ We can only partially teach intuitions (do not necessarily transfer across scales).
- ❑ Wrong interpretation: scale is all that matters, algorithms don't matter.
- ❑ Right interpretation: algorithms that scale is what matters.

$$\text{accuracy} = \text{efficiency} * \text{resources}$$

What are we going to cover in the class?

# Overview of the course (1)

- ❑ LLM basics
  - Attention and Transformer architecture
  - Decoder-only models (GPTs)
- ❑ Pre-training data and tokenization
- ❑ Architecture advancements on Transformers
- ❑ Evaluation
- ❑ Adaptation and learning
  - In-context learning, task-oriented finetuning, and parameter-efficient tuning methods
- ❑ Reasoning

# Overview of the course (2)

- ❑ Usage
  - RAG, Tool use
- ❑ Alignment
  - SFT, RLHF
- ❑ Scaling data and parameters
- ❑ Bias and ethics
- ❑ Efficient Pretraining and inference

# Overview of the course (3)

- ❑ LLM agents
- ❑ Long-context models
- ❑ Modularity
- ❑ Multilinguality and crosslingual transfer
- ❑ Vision language models
- ❑ Interpretability: probing, editing, induction heads

# Teaching Assistants

- ❑ Ali Ramezani ([ali.ramezani.96@ut.ac.ir](mailto:ali.ramezani.96@ut.ac.ir)) (Chief TA)
- ❑ Ali Khorramfar ([khoramfar@ut.ac.ir](mailto:khoramfar@ut.ac.ir))

# Prerequisites

- ❑ NLP course (**mandatory**)
- ❑ Deep Learning course
- ❑ Python Programming

# Grading

- ❑ Assignments: 30%
- ❑ Final project: 20%
- ❑ Exams:
  - Midterm: 20%
  - Final: 25%
  - Pop quizzes: Up to 10%

# Final project

- ❑ Choose a topic
- ❑ Project proposal
- ❑ Building teams (2-3)
- ❑ Progress report (initial phases of a paper)
- ❑ Final report / paper / presentation

# Project types

## ❑ Application

- A new model or scenario of using LLMs in a task
- The task could be in different fields, e.g., IR, vision, psychology, sociology, and politics

## ❑ Analysis

- Finding an unknown issue with LLMs, or analyzing the outputs/representations

This project can be part of a project which you've already been involved in or an initiation for your dissertation.

# Exams

## □ Exam dates:

- **Midterm exam:** During the class on 12/2/1405 (۱۲ اردیبهشت ۱۴۰۵ در زمان کلاس)
- **Final exam:** From 8:30 to 11:30 on 23/4/1405 (صبح ۲۳ تیر ۱۴۰۵)

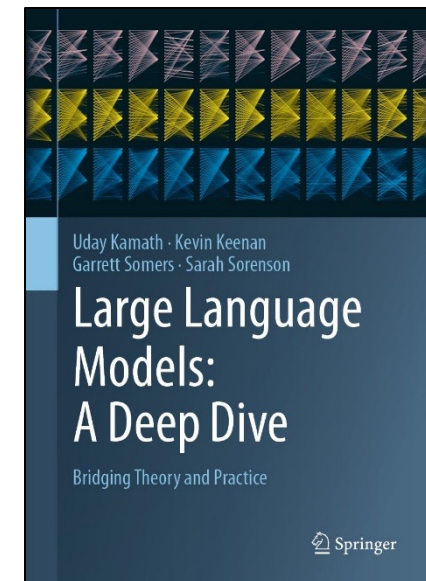
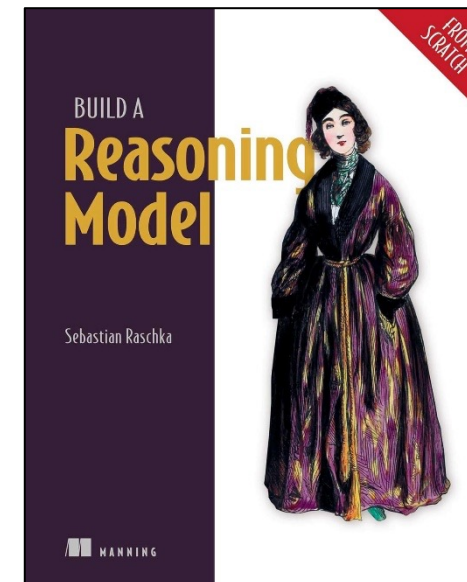
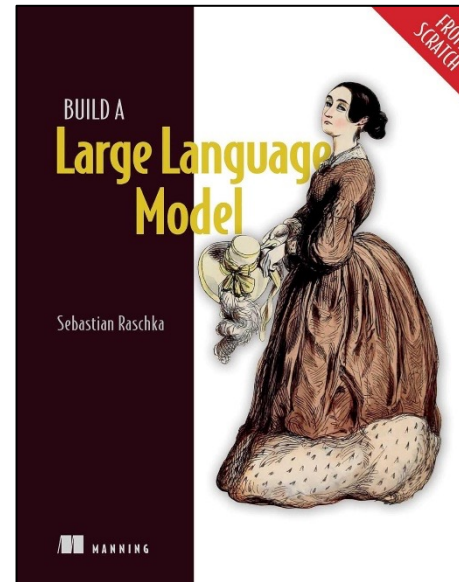
# Readings

## ❑ Main References:

- Course handouts.
- Assigned reading papers.

## ❑ Recommended Readings:

- W. X. Zhao et al. “A survey of large language models.” arXiv preprint arXiv:2303.18223 1.2 (2023): 1-124.



# LLM Courses from Top Universities

## ❑ CMU

- [Large Language Model Applications](#)

## ❑ Stanford

- [CS336: Language Modeling from Scratch](#)
- [CME 295 - Transformers & Large Language Models](#)

## ❑ Princeton

- [Deep Dive into Large Language Models](#)
- [Large Language Model Systems](#)

## ❑ University of Maryland, College Park

- [Advanced Natural Language Processing](#)

# Grading & Course Policies

- ❑ You are responsible for all assigned readings and information presented in the class, including due dates, assignments, exams and so forth.
  - Make sure to always download latest copies of course slides. I may revise them over time with more up-to-date info or fix errors over time. I use a versioning system by attaching **\_rev#** postfix to slide names and add a **changelog** slide to the very beginning of each slide deck.
- ❑ **You are expected to attend *all* classes.**
  - I won't take notes of absentees but you will be responsible for anything said verbally or done in the class.
  - Also, you need to attend classes in order to take quizzes.

# Educational Fairness

- ❑ We have a **zero-tolerance policy** for any form of **plagiarism** in this course.
  - Plagiarism: The practice of taking someone else's work or ideas and passing them off as one's own.
    - Needless to say, this includes copying solutions or codes from the Internet.
    - AI-assisted problem solving: If you get help from any AI tool (such as ChatGPT, Gemini, Copilot, etc.) to solve a problem, **you must include all prompts you used to reach to the solution AND a link to your query.** Failure to do so is considered plagiarism. Moreover, partial grades will not be granted to AI-assisted solutions.
  - First and foremost, note that plagiarism makes the course unfair to others.
  - You'll pickup a BAD habit which will hunt you down at some point in your life.
  - For the first time, you'll get a zero in the related work and for the second time, you will fail the course with the **lowest** grade.
  - **All parties** involved in plagiarism will be punished.
  - I really discourage it in any form. It isn't really worth it. You have been warned! 🦴

Share

# Educational Fairness (cont'd)

## ❑ Extreme situations:

- If you need any extra accommodations, feel free to raise your concern during my office hours or through email.
- I'll do my best to provide *reasonable* accommodation as long as the concern is raised in a timely manner.
  - Do **NOT** wait until the end of semester or when you get a bad grade. In this case, you probably won't get any accommodation.